What Kind of Model Is WOFOST?

The math behind the crop simulator, step by step

Gradient Descent Farmers — companion to the technical report

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1. Short answer

WOFOST is **not a regression**. It is a **process-based (mechanistic), dynamic, deterministic simulation model**: a system of ordinary differential equations that describes how a plant grows from physical and physiological processes (photosynthesis, respiration, water uptake), solved forward in time one day at a time. It never sees observed yields; the yield is what comes out at the end.

	Regression / machine learning	**WOFOST (process-based)**
Form	$y = \beta_0 + \beta_1 P + \beta_2 T + \varepsilon$	$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta})$, 20+ coupled equations
Numbers come from	Fitted to past yields ($\hat{\beta} = \arg\min \sum (y - \hat{y})^2$)	Measured plant physiology (leaf photosynthesis, respiration costs…)
Uses observed yields?	Yes, many	No (only optionally, to calibrate)
Explains *why*?	No, only correlations	Yes: every intermediate quantity can be inspected
New conditions (e.g. 40% less rain)	Unreliable outside the data	The same physics still applies
Main weakness	Needs lots of yield data	Many parameters; needs calibration to be accurate locally

2. The model as a dynamical system

State. At day t the field is described by a state vector

$$\mathbf{x}(t) = \big(\underbrace{D}_{\text{stage}},\ \underbrace{W_{LV},\, W_{ST},\, W_{RT},\, W_{SO}}_{\text{leaves, stems, roots, grain (kg/ha)}},\ \underbrace{\text{LAI}}_{\text{leaf area}},\ \underbrace{W,\, W_{low}}_{\text{soil water (cm)}},\ \underbrace{N_{av},\, N_{LV}, \ldots}_{\text{nitrogen (kg/ha)}} \big). \tag{1}$$

Dynamics. Each state changes at a rate that depends on the current state, today's weather $\mathbf{u}(t)$ (temperature, radiation, rain, humidity, wind) and fixed crop and soil parameters $\boldsymbol{\theta}$:

$$\frac{d\mathbf{x}}{dt} = f\big(\mathbf{x}(t),\, \mathbf{u}(t),\, \boldsymbol{\theta}\big). \tag{2}$$

Solution. The equations are solved numerically with the forward Euler method and a time step of one day:

$$\boxed{\mathbf{x}_{t+1} = \mathbf{x}_t + f(\mathbf{x}_t,\, \mathbf{u}_t,\, \boldsymbol{\theta}) \cdot \Delta t, \qquad \Delta t = 1 \text{ day}} \tag{3}$$

Each day: compute all rates from today's state and weather, then add them to get tomorrow's state.

Yield. The yield is the grain pool on the last day, i.e. the sum of daily grain growth:

$$Y = W_{SO}(t_{\text{end}}) = \sum_{t=t_{\text{sow}}}^{t_{\text{end}}} \Delta W_{SO}(t). \tag{4}$$

Written as one function, the whole season is $Y = \Phi(\mathbf{x}_0, \mathbf{u}_{1:T}, \boldsymbol{\theta})$: a deterministic composition of T daily updates. The same weather path always gives the same yield; all variation between years comes from the weather.

3. One simulated day, equation by equation

Symbols: T mean temperature (°C), I absorbed light, LAI leaf area index (m^2 leaf per m^2 ground), θ soil moisture (m^3/m^3). Functions written $g(\cdot)$ are piecewise-linear lookup tables of measured values.

Step 1 — Development (the crop's clock)

$$\frac{dD}{dt} = \frac{g(T)}{\text{TSUM}_k}, \qquad g(T) = \begin{cases} 0, & T \le 6\,^\circ\text{C} \\ T - 6, & 6 < T < 30 \\ 24, & T \ge 30 \end{cases} \quad \text{(maize)}, \tag{5}$$

with $\text{TSUM}_1 = 695$ °C d before flowering ($D < 1$) and $\text{TSUM}_2 = 860$ °C d after ($1 \le D < 2$). $D = 2$ is maturity. Water and nitrogen stress do not slow this clock.

Step 2 — Light through the canopy (Beer's law)

$$I(L) = I_0\, e^{-kL}, \tag{6}$$

where L is the leaf area above a given leaf and $k \approx 0.6$. Upper leaves get more light than lower ones.

Step 3 — Photosynthesis of one leaf (saturating curve)

$$A(I) = A_{\max}\left(1 - e^{-\varepsilon I / A_{\max}}\right), \tag{7}$$

linear in dim light (slope ε), levelling off at $A_{\max}$ in bright light. In WOFOST 8.1, $A_{\max}$ depends on the nitrogen in the leaf:

$$A_{\max} = f_{CO_2}\, f_T(T_{day})\, \min\big(A_{\text{ref}},\ s \cdot \text{SLN}\big), \tag{8}$$

with SLN the leaf nitrogen per unit leaf area. Less nitrogen means a lower ceiling.

Step 4 — Whole canopy, whole day (numerical integration)

The daily total integrates A over canopy depth and over the hours of daylight. WOFOST approximates both integrals with 3-point **Gaussian quadrature**:

$$\int_0^1 h(x)\,dx \ \approx\ \sum_{i=1}^{3} w_i\, h(x_i), \tag{9}$$

$$x = \left(\tfrac{1}{2} - \tfrac{\sqrt{15}}{10},\ \tfrac{1}{2},\ \tfrac{1}{2} + \tfrac{\sqrt{15}}{10}\right) \approx (0.113,\ 0.5,\ 0.887), \qquad w = \left(\tfrac{5}{18},\ \tfrac{8}{18},\ \tfrac{5}{18}\right), \tag{10}$$

which is exact for polynomials up to degree 5. Applied twice (3 times of day × 3 depths):

$$G = \mathrm{DAYL} \sum_{i=1}^{3} w_i \ \mathrm{LAI} \sum_{j=1}^{3} w_j \, A\big(I_{ij}\big) \quad [\text{kg } CO_2\text{/ha/day}]. \tag{11}$$

These are the only "weights" in WOFOST: numerical integration weights, not regression coefficients.

Converted to sugar, with a penalty for cold nights:

$$P = G \cdot g_{\min}\big(\bar{T}_{\min,\,7d}\big) \cdot \frac{30}{44} \quad [\text{kg } CH_2O\text{/ha/day}], \tag{12}$$

where 30/44 is the ratio of the molar masses of CH_2O and CO_2, and $g_{\min}$ is 0 at ≤ 5 °C and 1 at ≥ 8 °C.

Step 5 — Water stress

$$T_{\max} = ET_0 \left(1 - e^{-0.75\,k\,\mathrm{LAI}}\right) \qquad \text{maximum transpiration} \tag{13}$$

$$\theta_{cr} = (1-p)\,(\theta_{fc} - \theta_{wp}) + \theta_{wp} \qquad \text{moisture where stress starts} \tag{14}$$

$$f_w = \mathrm{clip}\left(\frac{\theta - \theta_{wp}}{\theta_{cr} - \theta_{wp}},\ 0,\ 1\right) \qquad 1 = \text{no stress, } 0 = \text{no water} \tag{15}$$

$$P_a = f_w \cdot P, \qquad T_a = f_w \cdot T_{\max} \qquad \text{actual photosynthesis and transpiration} \tag{16}$$

θ_{fc} is field capacity, θ_{wp} wilting point, and p depends on the drought sensitivity of the crop and on ET_0. ET_0 itself comes from the Penman–Monteith equation (sunlight, air dryness, wind).

Step 6 — Respiration (running costs)

$$M = \big(r_R W_{RT} + r_L W_{LV} + r_S W_{ST} + r_O W_{SO}\big) \cdot Q_{10}^{(T-25)/10}, \qquad Q_{10} = 2, \tag{17}$$

i.e. each kg of tissue costs a little sugar per day, doubling with every 10 °C.

Step 7 — Growth and partitioning

$$\Delta W = C\,(P_a - M), \qquad C \approx 0.67 \text{ kg dry matter per kg sugar}, \tag{18}$$

$$\Delta W_{RT} = F_R\,\Delta W, \qquad \Delta W_{LV} = F_L(1-F_R)\,\Delta W, \tag{19}$$

$$\Delta W_{ST} = F_S(1-F_R)\,\Delta W, \qquad \Delta W_{SO} = F_O(1-F_R)\,\Delta W, \tag{20}$$

where the fractions F_R, F_L, F_S, F_O depend on the stage D: roots and leaves early, almost everything to the grain after flowering ($F_O = 1$ from $D = 1.2$ in maize). New leaf area is $\Delta\mathrm{LAI} = \mathrm{SLA}(D)\,\Delta W_{LV}$ minus leaves dying of age, shading, drought or lack of N.

Step 8 — Soil water (mass balance)

$$W_{t+1} = W_t + \underbrace{R_{in}}_{\text{rain infiltrating}} - \underbrace{T_a}_{\text{crop}} - \underbrace{E_s}_{\text{soil evaporation}} - \underbrace{Q}_{\text{drainage}}, \qquad \theta = W / R_{\text{root}}. \tag{21}$$

Step 9 — Nitrogen (WOFOST 8.1)

$$D_N = \sum_{\text{organs}} \Big[\max\big(n^{\max} W - N,\, 0\big) + n^{\max}\, \Delta W \Big] \qquad \text{demand} \tag{22}$$

$$F_N = \phi\, D_N \qquad \text{fixed from the air} \tag{23}$$

$$U_N = \begin{cases} \min\big(D_N - F_N,\ N_{av},\ U_{\max}\big), & f_w > 0.01 \\ 0, & \text{otherwise} \end{cases} \qquad \text{taken from the soil} \tag{24}$$

$$N_{av,\,t+1} = N_{av,\,t} + \eta\, N_{\text{fert}} + \text{mineralization} - U_N \qquad \text{soil pool} \tag{25}$$

Faba bean has $\phi = 0.8$; maize and barley $\phi = 0$. A dry crop takes up no nitrogen, which links drought and nitrogen stress.

All steps in one line

Substituting steps 3–7 into the grain equation shows how one day of grain growth is built:

$$\Delta W_{SO} = \underbrace{F_O(1 - F_R)}_{\text{share to grain}} \cdot \underbrace{C}_{\text{sugar}\to\text{tissue}} \cdot \Big[\underbrace{f_w}_{\text{water}} \cdot \underbrace{\text{DAYL} \sum_i w_i\, \text{LAI} \sum_j w_j\, A(I_{ij}) \cdot g_{\min} \cdot \tfrac{30}{44}}_{\text{photosynthesis}} - \underbrace{M}_{\text{respiration}} \Big]. \tag{26}$$

4. One real day, by hand

Maize after barley at Texcoco, 7 August 2023 (WOFOST 7.2 core equations, which WOFOST 8.1 shares except for the leaf-nitrogen term). Every value below was recomputed independently and matched the model to all printed decimals.

Quantity	Calculation	Value
ET_0	Penman–Monteith: 3.996 (radiation) + 1.112 (dry air, wind)	5.108 mm
dD/dt	$g(17.56)/860 = 11.56/860$	0.013442
G	12.78 h $\times$ $(0.278 \cdot 109.8 + 0.444 \cdot 84.3 + 0.278 \cdot 21.1)$	943.79 kg CO_2/ha
P	$943.79 \times 1 \times 30/44$	643.49 kg CH_2O/ha
f_w	$(0.2225 - 0.137)/(0.2303 - 0.137)$	0.916
P_a	643.49×0.916	589.43 kg CH_2O/ha
M	$232.50 \times 2^{(17.56-25)/10}$	138.82 kg CH_2O/ha
ΔW	$0.671 \times (589.43 - 138.82)$	302.36 kg/ha
ΔW_{SO}	$F_O = 1,\ F_R = 0$ at $D = 1.20$	**302.36 kg grain/ha**

Summing ΔW_{SO} over the 93 days of grain filling gives that season's yield: 17,700 kg/ha.

5. Where fitting *does* appear

1. **Behind the parameters.** Some relationships (e.g. how photosynthesis falls with temperature) were derived from experiments decades ago, partly by curve fitting. In a simulation they are fixed constants.
2. **Calibration** (not yet done). Adjust a few parameters $\boldsymbol{\theta}_c$ so simulated yields match observed ones:
$$\hat{\boldsymbol{\theta}}_c = \arg\min_{\boldsymbol{\theta}_c} \sum_k \Big(Y_k^{\text{obs}} - \Phi\big(\mathbf{x}_0,\, \mathbf{u}_k,\, \boldsymbol{\theta}_c\big) \Big)^2. \tag{27}$$
This is a least-squares problem, but over a simulator Φ, not a linear formula.
3. **Statistics on the outputs**: means, coefficients of variation, correlation with SMAP. They describe the results; they are not part of the model.

6. How it connects to machine learning

WOFOST is the *environment*; learning sits on top. In reinforcement learning (CyclesGym, CropGym), each season is a step of a Markov decision process:

$$s_t = \text{soil water, soil N, weather so far,} \qquad a_t = \text{crop, sowing date, fertilizer,} \tag{28}$$

$$s_{t+1} \sim \text{WOFOST}(s_t, a_t, \mathbf{u}_t), \qquad r_t = \text{price} \cdot Y_t - \text{cost}(a_t), \qquad \pi^* = \arg\max_{\pi} \; \mathbb{E}\Big[\sum_t \gamma^t r_t\Big]. \tag{29}$$

The crop model supplies physically realistic consequences; the learning algorithm finds the decisions.